

2. The *FileWriterUtil* is the originator class, contains 2 fields: filename, content. The *FileWriterUtil* class contains a private inner class called *FileWriterUtilMemento*. The *write()* method of *FileWriterUtil* is used to update content. The *save()* method of *FileWriterUtil* creates and returns a *FileWriterUtilMemento* object. The *restore()* method of *FileWriterUtil* is used to restore the state of the *FileWriterUtil*. *FileWriterCaretaker* is the caretaker class that is used to store the state of a *FileWriterUtil* object. Finally, write the *Test* class to test the memento pattern.

FileWriterCaretaker

```
package experiment19_2;

class FileWriterCaretaker {

    private FileWriterUtil.FileWriterUtilMemento memento;

    public void saveState(FileWriterUtil fileWriterUtil){

        memento = fileWriterUtil.save();

    }

    public void restoreState(FileWriterUtil fileWriterUtil){

        fileWriterUtil.restore(memento);

    }

}
```

FileWriterUtil

```
package experiment19_2;

class FileWriterUtil {

    private String filename;

    private String content;
```

```
public FileWriterUtil(String filename){
    this.filename = filename;
    this.content = "";
}

public void write(String content){
    this.content = content;
}

public FileWriterUtilMemento save(){
    return new FileWriterUtilMemento(filename,content);
}

public void restore(FileWriterUtilMemento
fileWriterUtilMemento){
    this.filename = fileWriterUtilMemento.filename;
    this.content = fileWriterUtilMemento.content;
}

public String getFilename(){
    return filename;
}

public String getContent(){
    return content;
}

class FileWriterUtilMemento{
```

```
    private String filename;

    private String content;

    public FileWriterUtilMemento(String filename, String
content){

        this.filename = filename;

        this.content = content;

    }

    private String getFilename(){

        return filename;

    }

    private String getContent(){

        return content;

    }

}

}
```

Test

```
package experiment19_2;

public class Test {

    public static void main(String[] args) {
```

```
        FileWriterCaretaker fileWriterCaretaker = new
FileWriterCaretaker();

        FileWriterUtil fileWriterUtil = new
FileWriterUtil("testFile.txt");


        fileWriterUtil.write("Hello World!");

        System.out.println("Current Content: " +
fileWriterUtil.getContent());


        fileWriterCaretaker.saveState(fileWriterUtil);


        fileWriterUtil.write("Hello World again!");

        System.out.println("Updated Content: " +
fileWriterUtil.getContent());


        fileWriterCaretaker.restoreState(fileWriterUtil);

        System.out.println("After restoration, Content: " +
fileWriterUtil.getContent());

    }
}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ
Current Content: Hello World!
Updated Content: Hello World again!
After restoration, Content: Hello World!

Process finished with exit code 0
```